

Biological and climatic constraints drive reproductive synchrony

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Abstract

Synchronous ‘boom-and-bust’ reproduction—or masting—is common across tree species globally. Decades of research have identified climate as a potential driver of the timing of masting, raising concerns anthropogenic climate change may disrupt it. Yet exactly how trees reproduce synchronously—a critical component of masting—is harder to explain. Here, using long-term seed production data from individual beech trees across England, we scale up tree-level predictions to population-level dynamics, to show how climate and biological constraints together drive reproductive synchrony. Our approach integrates climatic cues with a reproductive constraint within each tree: overlap between floral bud initiation and fruit maturation that drives a trade-off between the two processes across years. Our results support the idea of masting as a synchronous ‘boom-and-bust’ cycle, with 87% of trees within each population sharing the same reproductive state in any given year. Synchrony between populations did not decline significantly over the last decade despite marked warming (84% before 2015 versus 83% after). Warm summers increased the probability that trees transitioned into a high reproductive state. Individual-level reproductive constraints, however, prevented runaway increases in masting frequency, even under forecasts of extreme warming. We found synchrony was driven mainly by cold summers, which pushed most trees into a shared low reproductive state, providing the conditions for synchronous reproduction when a warm summer followed. Our results suggest the current interplay of individual-level constraints and population-level climatic cues will lead to continued synchrony, and highlight how individual-level data combined with mechanistic models can improve large-scale forecasts.

Introduction

The acceleration of climate change may cause abrupt ecological effects worldwide¹. Increasingly extreme conditions could disrupt key ecological processes, and drive ecosystems toward critical transitions². Recent work has suggested major shifts in pests and pathogens across biomes, while the local extirpation of rare species and increase in invaders has driven biotic homogenization^{3–5}. Forests in particular appear to be shifting, with lagging elevational and latitudinal range expansions, alongside increases in adult tree mortality⁶. This has driven growing concerns about the future of forests to act as carbon sinks globally^{7,8,6}, and thus growing pressure to understand the mechanisms that underlie forest regeneration^{9–12}.

Regeneration in many temperate and tropical forests depends on the widely observed phenomenon of masting. Masting—defined by ‘boom and bust’ cycles of reproduction that are synchronous across most individuals of a population¹³—occurs across many tree species and is hypothesized to drive fitness benefits during reproduction and the early stages of establishment^{14,15}. The most commonly invoked fitness benefit, ‘predator satiation,’ hypothesizes that years of high seed production overwhelm seed predators and thus allows a higher proportion of seeds and seedlings to escape predation and establish^{16,14}. Fitness benefits of masting may also operate earlier in reproduction, by increasing pollen exchange and genetic outcrossing^{17,18}. While these different hypotheses vary in their exact mechanism and lifestage, they all imply that the benefits of masting depend on an economy of scale: the benefits increase as more trees within a population (or location) reproduce synchronously.

Despite the importance of synchrony to yielding fitness benefits, how synchronous masting is and how it occurs remain poorly understood. Most hypotheses for how synchrony emerges focus on shared environmental cues at the species-level^{19,15}. Among many potential cues, an abundance of work suggests that climatic conditions during bud formation may lead to masting^{20,21}. In temperate systems, sufficiently warm temperatures during the growing season before masting—when reproductive buds generally develop—alongside mild spring conditions that do not cause tissue loss after buds form (i.e. no late spring frosts^{22,23}) appear to drive masting. Climate change, however, is rapidly shifting temperatures across the spring and summer^{24–26}, and thus could disrupt masting dynamics. Such changes would have major implications for forest resilience^{27,9}, but forecasting them requires better understanding how these climatic cues affect reproductive biology, from buds to fruits.

Most temperate woody species develop reproductive buds through a multi-year process. In the common two-year reproductive cycle (Fig. 1A-B), bud formation begins the year before flowering and seedset—potentially explaining why summer temperatures from the previous year appear to trigger masting^{28,29}. The following year some of these buds may be lost to spring frost, potentially reducing the number of seeds in what otherwise would have been a masting year. This two-year cycle, however, also introduces a constraint in most species (Fig. 1C). Because floral buds for the following year develop simultaneously with current-year fruit²⁸, a large fruit crop can depress bud initiation (through hormonal inhibition and within-plant resource competition^{30–32}). The result is a physiological constraint on flower and fruit development that is often invoked to explain why many woody species show alternate bearing^{30,29}, with a large crop production year (‘on-year’) often followed by one or several years of low production (‘off-years’). Effectively, these constraints can be conceptualized as producing latent reproductive states (low or high) that shape seed production in the following year (observed as off- and on-years in crops).

These reproductive constraints and states may shape how climate—and climate change—impacts seed production. Ideal climatic conditions should trigger higher investment in floral buds only for trees not already in a high reproductive state (Fig. 1D). Warm summers would therefore not affect all trees equally: trees in a high reproductive state should be largely unaffected, while those in a low reproductive state should become more likely to transition to a high state. Once a reproductive

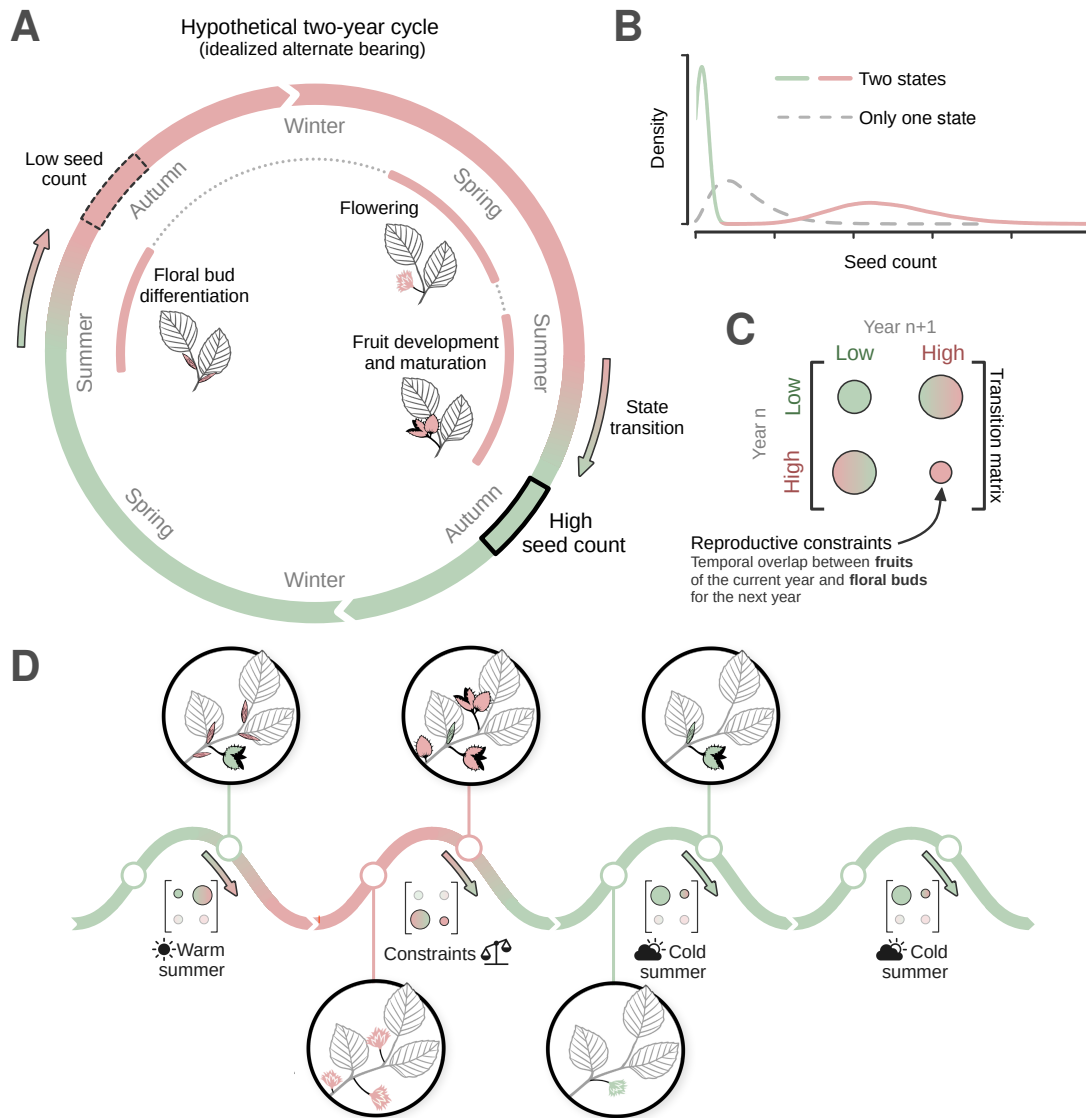


Figure 1: **A two-state model captures the interplay of reproductive constraints and climatic cues.** **(A)** A large investment in floral buds in summer (initiation and differentiation) leads to a large flowering the following spring (high reproductive state shown in red, transitions shown with arrows). Under favorable conditions, many fruits then develop and mature, producing a high seed count in autumn. Investment in the fruits reduces the investment in floral buds, leading to a low seed count the year after (low reproductive state shown in green). **(B)** High (red) and low (green) reproductive states lead to a different distribution of seed counts, respectively, which a single-state model (gray) would not explain. **(C)** Transitions between low and high reproductive states are governed by a matrix of transition probabilities. The probability of staying in a high state is likely low, reflecting the reproductive constraint: the temporal overlap between maturing fruit in the current year and floral bud formation for the next year. **(D)** On top of this constraint, climate, and summer temperature in particular, influence the transition probabilities. A warm summer may push a tree into a high reproductive state, but the tree is unlikely to stay in a high state the next year because of the reproductive constraint.

state has been established, spring frost may matter, with damage from frosts most likely to reduce potential seed production in trees in a high state. Together, these individual-level reproductive dynamics could scale up to generate population-level synchrony, with mastings emerging through a combination of climatic variability and reproductive constraints.

Here we investigate how climate affects mastings by explicitly modeling how climatic cues interact with individual reproductive constraints to generate population-level synchrony. Using seed count data collected from individual trees across populations of beech forests of England since 1980 (Supp. Fig. S1), we first test whether individual trees show evidence for alternative reproductive states. We then examine how synchrony arises at the individual level and scales up to populations. Finally, we leverage these results to forecast how future warming may affect both tree-level seed production and population-level synchrony.

Results and discussion

Following the paradigm that mastings is defined by of ‘boom’ and ‘bust’ reproduction, our model identified high and low reproductive years as alternative latent states (Fig. 2). In high (‘boom’) reproductive years, the model estimates 168 seeds on average (90% uncertainty range: 163 – 173 seeds counted on the ground beneath an individual tree, see methods), while in low (‘bust’) reproductive years they produced 52 seeds on average (48 – 56 seeds per tree). The distribution of seed count per tree in low and high reproductive years, however, overlapped (Fig. 2A). The difference between these two states occurred mostly in the opposite extremes: years in the high reproduction latent state included very high seed counts, which would almost never be possible in the low reproduction state, while years in the low reproduction state were characterized by extremely low counts, including a high probability of no observed seeds at all (Fig. 2A, Supp. Fig. S2).

Transitioning between reproductive states (high or low) depended on a tree’s state in the previous year (Fig. 2B). In a year with average climatic conditions (average max. summer temperatures of 19.7 °C), a tree in a low reproductive state was most likely to transition to a high reproductive state the next year, with a probability of 64% (61 – 68%), and a 1.8 times lower probability (36%, 32 – 39%) of staying in a low reproductive state (Fig. 2B). Under the same average climate, a tree in a high reproductive state was almost certain to transition to a low reproductive state, with a probability of 99% (97 – 100%), and very unlikely to remain in the same state (1%, 0 – 3%). Trees thus commonly shift into the opposite reproductive state each year, but the low reproductive state is much more persistent (Fig. 2C-D). Conversely, trees rarely stay in the high reproductive state for more than a single year, supporting the idea of non-mast years being more frequent than mast years¹⁴.

Climate strongly affected reproductive state transition dynamics. Similar to previous studies^{20,22,33}, we found that warm summer temperatures increased the probability that trees in a low reproductive state transitioned to a high reproductive state (Fig. 3A, Supp. Fig. S3). For example, the transition probability during a warm summer at 22 °C was 3.6 times higher than during a 18 °C summer (94% vs. 26% respectively, and ranges of 92 – 96% vs. 22 – 30%, Supp. Fig.

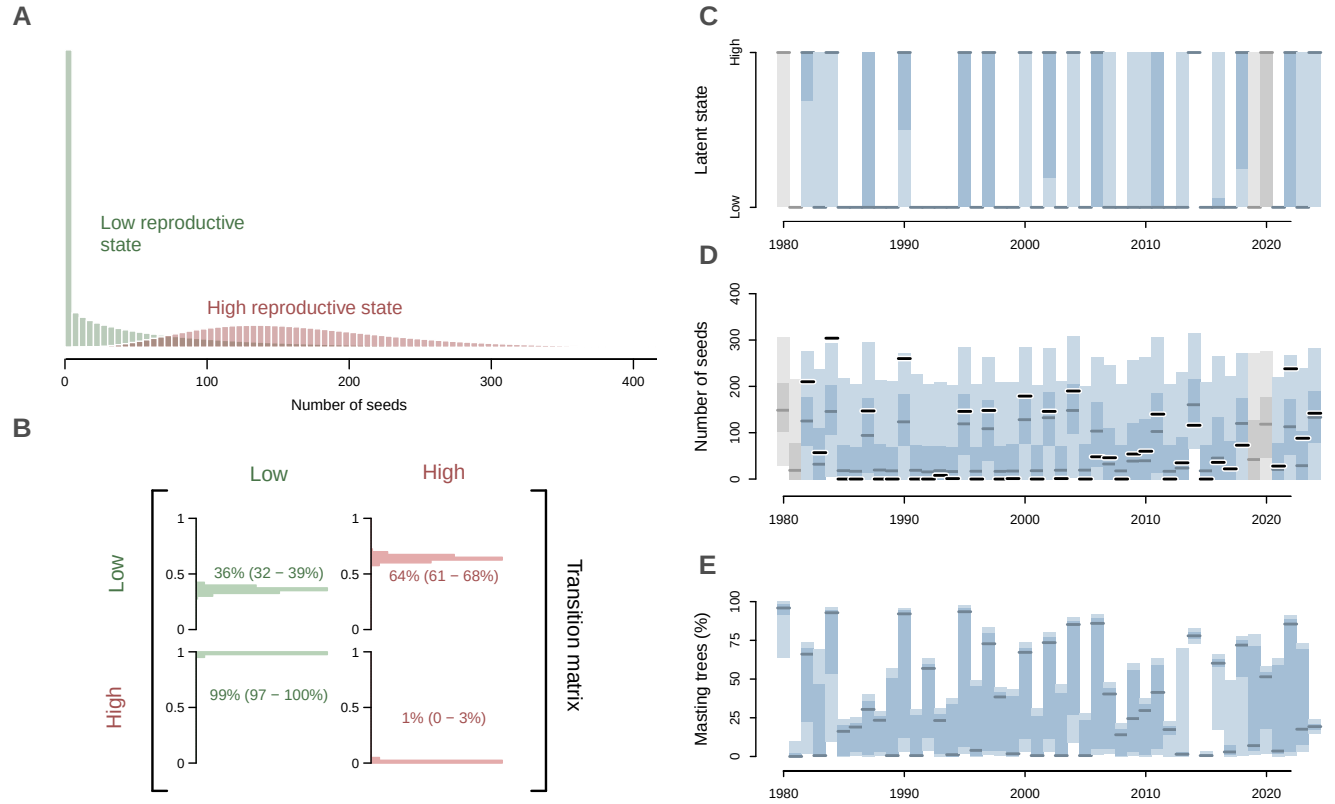


Figure 2: **Distinct tree reproductive states generates variability at the population level.** **(A)** Our two-state model (see Fig. 1) identifies two distinct reproductive latent states (high and low) that correspond to two different levels of seed production. **(B)** Differences in transition probabilities between the two latent states reflect constraints that shape tree-level reproductive cycles. **(C-D)** At the tree level (illustrated here for one arbitrary tree), these alternating latent states over time **(C)** generate low and high seed production years **(D)**. **(E)** When added together across all populations (Supp. Fig. S1), these individual cycles generate variability and synchrony in seed production across years. In **(C)**, **(D)** and **(E)**, we show median model predictions (blue lines) with 50% and 90% uncertainty intervals (shaded areas) overlaid with observed seed counts (black lines) or model predictions for years when observations were missing (gray).

S4). Additionally, our results support the common hypothesis that spring frosts may limit seed production^{34,35}, with seed production for trees in a high reproductive state reduced by 20% (14 – 26%, Fig. 3B) given a three-fold increase in a metric of frost risk (growing-degree days until last frost³⁶), but no evidence that warm springs affected seed production (Supp. Fig. S2). Given the reproductive constraints that prevent beeches and many other woody plants from repetitive years of high fruit (from the competitive overlap of flower bud formation and fruit development^{30,28}), we assumed the transition from a high to a low reproductive state was not affected by climate (we found strong support for this when we relaxed this assumption, Supp. Fig. S5).

Individual tree-level reproductive states (high or low) scaled up to produce population-level seed production that was synchronous, both within and between populations (forest sites). Within populations, 87% (84 – 89%) of trees shared the same reproductive state on average each year, while synchrony across populations was only slightly lower (the proportion of trees in the same state across all populations was 84%, 81 – 86%). This high level of synchrony across populations could suggest that synchrony occurs across wide spatial scales, as sometimes suggested^{37,33}. However, because these populations were not widely separated spatially (Supp. Fig. S1), these results only provide evidence for synchrony over smaller spatial scales, as often suggested^{38,39}. Whether this scale is sufficient for fitness benefits depends on the exact mechanism (e.g., predator satiation or pollination efficiency) and the species-level attributes (e.g. predators foraging ranges or wind dispersal distances, which depend on seed shape, size and the local landscape⁴⁰). For example, if beech masting in England mainly serves to overwhelm seed predators, such as wood mouse or bank vole⁴¹, then the masting synchrony at the population level that we observe should be sufficient to maintain the regeneration benefits of masting (given that sites are on average 0 km apart, and the two closest sites 0 km apart).

Climatic cues—especially cold summers—appeared critical to synchronizing the reproductive cycle of individual trees to produce the population-level synchrony that often defines masting¹⁹. While 36% of trees in a low reproductive state are likely to stay in that state, cold summer temperatures reduce the probability that an additional subset of trees transitions to a high reproductive state (Fig. 3A). Thus, reproductive constraints combined with cold summers can leave most trees in a low reproductive state. This synchrony of low reproduction appears to then drive synchrony of high reproduction when a cold summer is followed by a warm summer. In such years, most trees responded consistently to the warm temperatures that promote floral induction, and transitioned together into a high reproductive state (Fig. 3C), generating a mast year at the population scale. This interplay between individual-level constraints and population-level climatic cues provides a physiological mechanism for reproductive synchrony that can persist over multiple years, and offers a simple explanation for previous findings that masting depends on the temperature difference between successive summers (the ΔT model⁴²).

Our results appear to support the hypothesis that increasing summer temperatures with climate change could disrupt masting^{43,9,10}. This hypothesis—previously called breakdown—predicts that warm summers will erode both the inter-annual variability of individual tree reproduction and the synchrony of reproduction within populations, i.e. the two components of masting. With warmer

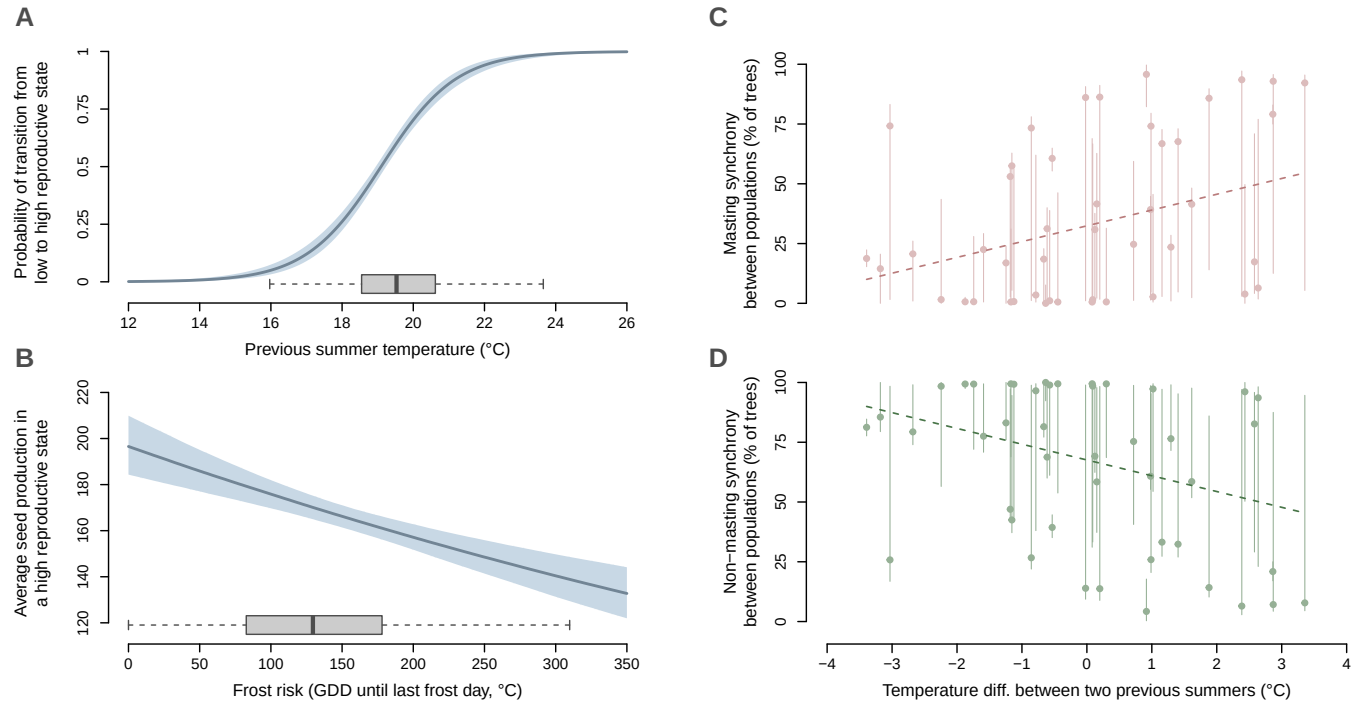


Figure 3: **Climate conditions impact both transition probabilities between reproductive states and seed production.** (A) Average maximum temperature in July and August of the previous year increases the probability of transition from a low to high reproductive state. (B) Frost risk (measured as accumulated growing-degree days, GDD, until the last frost day³⁶) decreases the number of seeds produced when in a high reproductive state. (C-D) The interaction of previous-year summer conditions and latent reproductive states (C: high reproductive state, D: low reproductive state) explain the apparent two-year lag effect of climate on observed synchrony across populations. In (A) and (B), the blue line and shaded area respectively represent the median and the 90% uncertainty interval, while the gray boxplots show observed conditions (1980–2022). In (C) and (D), the dashed lines represent the mean prediction from linear regressions fitted to each posterior draw of synchrony.

152 summers, individual trees would invest in flowers and fruit in most years, potentially leading to
 153 lower tree growth^{38,10}. This loss of years of low reproduction would in turn reduce synchrony at
 154 the population level, minimizing the fitness benefits of masting⁴³. Because our model captures
 155 the reproductive states underlying observed seed production, it allows us to separate these two
 156 components of variability—at the inter-annual tree-level and at the population level (population
 157 synchrony)—and our results suggest that they respond differently to warming.

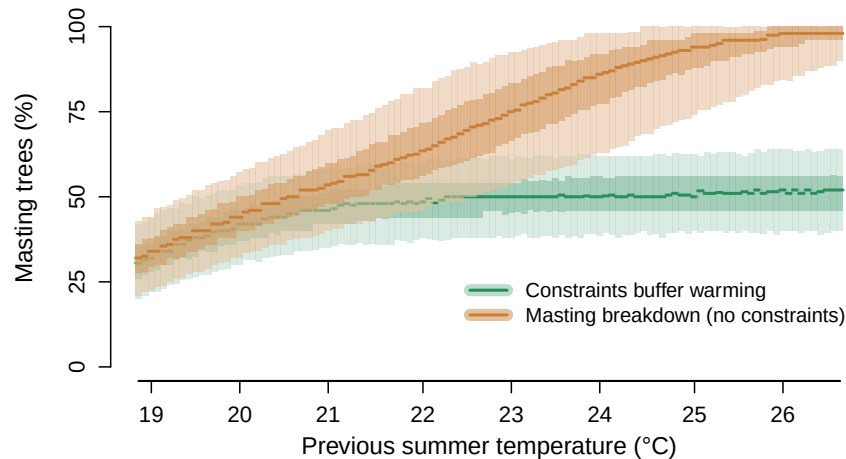


Figure 4: **Reproductive constraints prevent individual trees from reproducing in most years, even under significant summer warming.** Our results suggest trees cannot remain continuously in a high reproductive state with increased warming due to reproductive constraints (green). We could only replicate a potential breakdown of individual reproductive variability with warming through a hypothetical model (not supported by the data here) where we removed reproductive constraints and forced summer warming to increase the probability of persisting into a high reproductive state (brown, see Methods for more details). We show the medians (lines) with 50% and 90% uncertainty intervals (darker and lighter shading respectively) from model predictions.

158 For inter-annual variability, we find no evidence that warming pushes individual trees towards
 159 reproducing in most years. While we do find that warmer summers increase the probability of trees
 160 transitioning into a high reproductive state, and therefore increasing summer temperatures would
 161 increase the frequency of high seed production years, they do not increase the probability of a tree
 162 remaining in that state. Reproductive constraints continue to return trees to a low reproductive
 163 state, preventing trees from reproducing heavily in consecutive years, making negative effect of
 164 sustained reproduction on growth unlikely¹⁰. Indeed, we found that with extreme summer warming
 165 (7 °C in one century, corresponding to the warmest regional climate model projections²⁶), only 52%
 166 (40 – 64%) of trees in a population would mast in any given year (Fig. 4). This result holds when we
 167 relax reproductive constraints and allow warm summers to increase the probability of trees staying
 168 a high-reproductive state (Supp. Fig. S6). Even with this additional effect, we do not observe a
 169 breakdown of individual reproduction variability. On the contrary, we found a ‘breakdown’ because
 170 of climate change would require summer warming to have a similar impact on both the transition
 171 and persistence into a high reproductive year—an hypothesis not supported by the data used in
 172 this analysis (Fig. 4).

These results highlight how biological constraints on plant reproduction may limit runaway effects of climate change. For masting, floral bud initiation at the individual tree level is promoted by warm summer temperatures, potentially leading to a higher number of flowers and fruits in the following year^{22,15,33}. In the next summer, however, the development of a larger fruit load inherently imposes a trade-off that limits the tree from initiating a large number of floral buds again^{30,29}. These individual-level constraints scale up to the population level by preventing the entire population from producing large seed crops every year—and thus also preventing an unlimited amplification of climate effects on masting. Our results highlight that constraints at the individual level are critical for accurate forecasts of population-scale dynamics⁴⁴. Further this approach can help identify the spatial scale at which changes occur, including possible shifts in synchrony.

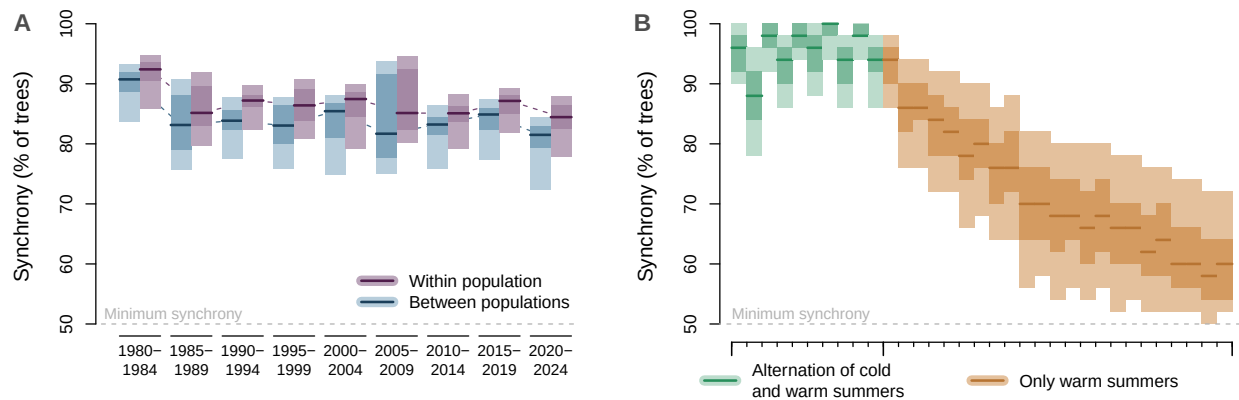


Figure 5: Reproductive synchrony has remained largely stable over recent decades, but persistent decline in cold summers could desynchronize tree reproduction in the future. (A) Observed synchrony within populations (purple, the average proportion of trees of the same population in the same reproductive state) and across populations (blue, the proportion of trees in the same state across all populations) has remained largely stable over recent decades. **(B)** In a hypothetical population, the synchrony is maintained while summers alternate between cold (around 17 °C) and warm (around 22 °C) conditions (green), but declines once the population experiences only warm summers (orange). X-axis shows hypothetical years. In **(A)** and **(B)**, we show median model predictions (solid lines) with 50% and 90% uncertainty intervals (shaded areas).

For population synchrony, we found no clear evidence of declines in synchrony in recent decades, despite significant warming over this period^{45,9}. Synchrony across trees within a population varied over time but has not clearly changed in recent periods (recent declines in 2015 onward are well within the 90% range of previous periods, Fig. 5A). Similarly, synchrony between populations has been quite stable, from a previous mean of 84% (81 – 87%) before 2015 to one of 83% (79 – 85%) since 2015. This stability in synchrony across both scales suggests that masting in these beech populations may be more resilient to warming than previously predicted⁴³.

Whether masting can continue to support forest regeneration in the next decades, however, may depend on how synchrony responds as the climate continues to warm. While we find no evidence for a positive feedback of summer warming that would trigger runaway seed production at the individual level, our results predict possible changes at the population level in synchrony

as we lose the cold summers. Because cold summers act as potential hinge points for population synchronization, our results suggest that several consecutive years without cold summers decrease synchrony within a population (Fig. 5B). Current climate projections, however, suggest that such strings of warm summers remain rare for the next decades (Supp. Fig. S7; but climate models have underestimated recent summer warming in Western Europe²⁶). Determining the extent to which the ‘breakdown’ will happen in the coming decades, and producing robust spatiotemporal forecasts, will require better measurements of the temporal and spatial scales at which synchrony occurs³⁹. Our results show the value of building from individual-level seed production data, which is still incredibly rare, to understanding the population scale. Teasing out the full effects of warm summers versus spring frosts, however, will also require temporally finer resolution data than seed counts (e.g., flowering⁴⁶)—ideally following buds from formation to flowering to fruits to measure losses at each stage.

Fundamentally synchrony in seed production is one piece of a larger puzzle. Understanding future forest regeneration requires disentangling the full sequence of reproductive and growth phenology stages that lead from a floral bud to an established tree. New data on bud formation, flowering, and seed production across years could improve models of masting. Improving our understanding of this is the first step of a much longer sequence could help pinpoint how climate change will ultimately impact forest regeneration. Climate change may lead to major shifts in the later stages—seed germination, seedling establishment and tree recruitment⁴⁷, with potential global impacts on forests and the services they provide. Our results highlight that we may fail to accurately forecast these shifts until we integrate the biological and climatic constraints that start with individual trees.

Methods

Tree-level reproduction and climate data

We used annual seed production data from 171 individual beech trees (*Fagus sylvatica*) across 12 populations in England, collected between 1980 and 2024, following the protocol described in previous publications^{48,49,43,10}. Seed numbers used here represent nuts counted during seven minutes of ground searching under an individual tree, once a year usually in October^{48,49}. Populations contained between 8 and 21 trees and were monitored for 5 to 45 years. Within populations, individual time series varied in length (average length = 20.4 years) and contained some missing observations.

We extracted daily climatic data from ERA5-Land⁵⁰, at a 0.01 degree resolution, for the 12 sites. We computed three climatic variables for each year of observation: average maximum temperature in July and August (of the previous year), average spring temperature (April-May), and growing degree days (GDD) until the last frost day as a measure of spring frost risk³⁶.

A Hidden Markov model for individual tree reproduction

Structure of the model

We modeled individual tree seed production with a Hidden Markov Model (HMM), in a Bayesian framework using the probabilistic programming language Stan⁵¹. We present models here run on 4 chains of 2000 iterations each (including 1000 warm-up iterations) with no divergent transitions. For each parameter, we obtained a split- $\hat{R} < 1.01$ and a high empirical ESS (> 100 by chain). We provide all Stan code with the manuscript and will publish it on an open federated repository once accepted.

In this model, each tree k is assigned to a latent reproductive state $z_{k,i}$ in each year i : a low reproductive state ($z_{k,i} = 1$) or a high reproductive state ($z_{k,i} = 2$). The latent state $z_{k,i}$ is conditionally dependent on only the state the preceding year (the *Markovian* assumption), such that for N years the joint model is given by:

$$p(z_{k,1}, \dots, z_{k,N}) = p(z_{k,1}) \prod_{i=2}^N p(z_{k,i} | z_{k,i-1}) \quad (1)$$

The transitions between reproductive states are determined by a 2×2 matrix of probabilities:

$$\begin{bmatrix} p(z_{k,i} = 1 | z_{k,i-1} = 1) & p(z_{k,i} = 2 | z_{k,i-1} = 1) \\ p(z_{k,i} = 1 | z_{k,i-1} = 2) & p(z_{k,i} = 2 | z_{k,i-1} = 2) \end{bmatrix} \quad (2)$$

which is, by definition, equivalent to:

$$\begin{bmatrix} 1 - p(z_{k,i} = 2 | z_{k,i-1} = 1) & p(z_{k,i} = 2 | z_{k,i-1} = 1) \\ 1 - p(z_{k,i} = 2 | z_{k,i-1} = 2) & p(z_{k,i} = 2 | z_{k,i-1} = 2) \end{bmatrix} \quad (3)$$

Both the latent reproductive states and the transition probabilities were inferred from the seed data alone, i.e. the model identifies the most probable state sequence for each individual tree given its observed seed counts.

The latent state determines the observational model responsible for the observed seed count $y_{k,i}$. We modeled the seed production in a low reproductive state with a zero-inflated negative binomial distribution, and the one in a high reproductive state with a negative binomial distribution:

$$y_{k,i} | z_{k,i} = 1 \sim \begin{cases} 0 & \text{with probability } \theta \\ \text{NegBin}(\lambda_{\text{low}}, \psi_{\text{low}}) & \text{with probability } 1 - \theta \end{cases} \quad (4)$$

$$y_{k,i} | z_{k,i} = 2 \sim \text{NegBin}(\lambda_{\text{high}}, \psi_{\text{high}}) \quad (5)$$

Influence of climate on state transition and seed production

We expected the climatic conditions experienced by a tree to influence both the transition matrix (Equation 3) and the seeds produced in a high reproductive state (Equation 5). Conversely, we

assumed that climate does not impact seed production in a low reproductive state (Equation 4). In Equation 3, we assumed that only the transition from a low reproductive state to a high reproductive state was affected by climate (and thus, by construction, the probability of staying in a low reproductive as well):

$$p(z_{k,i} = 2 \mid z_{k,i-1} = 1) = \text{logistic}(\alpha_{1 \rightarrow 2} + \beta_{1 \rightarrow 2}^{\text{summer}} \cdot (X_{k,i-1}^{\text{summer}} - X_{\text{baseline}}^{\text{summer}})) \quad (6)$$

Here, $X_{k,i-1}^{\text{summer}}$ is the average maximum temperature during summer (the preceding year $i-1$). $\beta_{1 \rightarrow 2}^{\text{summer}}$ captures the impact of variations in summer temperature away from the baseline $X_{\text{baseline}}^{\text{summer}}$. Finally, $\text{logistic}(\alpha_{1 \rightarrow 2})$ is the transition probability when $X_{k,i-1}^{\text{summer}} = X_{\text{baseline}}^{\text{summer}}$. Because the temporal overlap between fruit development and floral bud initiation should prevent trees already in a high reproductive state from responding to favorable summer conditions, we did not expect $X_{k,i-1}^{\text{summer}}$ to affect $p(z_{k,i} = 2 \mid z_{k,i-1} = 2)$. We still verified that this effect was negligible (see Supp. Fig. S5 and S6).

In Equation 5, we assumed that both spring temperature $X_{k,i}^{\text{spring}}$ and spring frost risk $X_{k,i}^{\text{frost}}$ could impact the seed production in a high reproductive state, such that:

$$y_{k,i} \mid z_{k,i} = 2 \sim \text{NegBin}(\lambda_{k,i}^{\text{high}}, \psi_{\text{high}}) \quad (7)$$

$$\lambda_{k,i}^{\text{high}} = \exp\left(\log(\lambda_{\text{high}}) + \beta_{\text{high}}^{\text{spring}} \cdot (X_{k,i}^{\text{spring}} - X_{\text{baseline}}^{\text{spring}}) + \beta_{\text{high}}^{\text{frost}} \cdot (X_{k,i}^{\text{frost}} - X_{\text{baseline}}^{\text{frost}})\right) \quad (8)$$

Masting synchrony and model predictions

To quantify reproductive synchrony we aggregated individual tree states within and across populations, i.e. the proportion of trees sharing the same reproductive state (Fig. 3C-D, Fig. 5). Within populations, we computed synchrony as the maximum of the proportion of trees in a high or low reproductive state for each population (e.g. if 30% of the trees are in a low state and 70% in a high state, the synchrony is 70%), then averaged across populations. Across populations, we computed synchrony similarly, but we first averaged the proportions of trees in each state across all populations before taking the maximum (to avoid a population with more trees contributing more). Synchrony within populations can thus be higher than synchrony across populations when trees within a site tend to be in the same state but differ between sites. Since our model includes only two states, the minimum synchrony is 50% (i.e. at least 50% of the trees are in the same state). In Fig. 3C-D, we fitted a simple linear regression, using Stan, of masting synchrony on climate to each posterior draw of the synchrony estimates (from our HMM), and show the prediction given by the mean intercept and slope of those regressions to highlight the trend (see Supp. Fig. S8 for quantile intervals.)

To forecast the effects of summer warming (Fig. 4), we simulated seed production (and synchrony) for 50 trees representing an average population (from our data) with projected summer warming up to 7°C over one century²⁶. To assess the role of biological constraints, we compared predictions from our model—in which reproductive constraints prevent trees from staying indefinitely in a high reproductive state—against a hypothetical model where summer temperatures had a

similar effect on both $p(z_{k,i} = 2 \mid z_{k,i-1} = 1)$ (transition from low to high) and $p(z_{k,i} = 2 \mid z_{k,i-1} = 2)$ (persistence in high). To illustrate the role of cold summers in reproductive synchrony, we also simulated seed production for 50 trees first experiencing alternating cold (around 17 °C) and warm summers (around 22 °C), followed by a period of warm summers only (Fig. 5B).

References

- [1] Christopher H. Trisos, Cory Merow, and Alex L. Pigot. The projected timing of abrupt ecological disruption from climate change. *Nature*, 580(7804):496–501, April 2020. ISSN 1476-4687. doi: 10.1038/s41586-020-2189-9.
- [2] Thomas Wernberg, Scott Bennett, Russell C. Babcock, Thibaut de Bettignies, Katherine Cure, Martial Depczynski, Francois Dufois, Jane Fromont, Christopher J. Fulton, Renae K. Hovey, Euan S. Harvey, Thomas H. Holmes, Gary A. Kendrick, Ben Radford, Julia Santana-Garcon, Benjamin J. Saunders, Dan A. Smale, Mads S. Thomsen, Chenae A. Tuckett, Fernando Tuya, Mathew A. Vanderklift, and Shaun Wilson. Climate-driven regime shift of a temperate marine ecosystem. *Science*, 353(6295):169–172, July 2016. ISSN 1095-9203. doi: 10.1126/science.aad8745.
- [3] Colin J. Carlson, Cole B. Brookson, Daniel J. Becker, Caroline A. Cummings, Rory Gibb, Fletcher W. Halliday, Alexis M. Heckley, Zheng Y. X. Huang, Torre Lavelle, Hailey Robertson, Amanda Vicente-Santos, Ciara M. Weets, and Timothée Poisot. Pathogens and planetary change. *Nature Reviews Biodiversity*, 1(1):32–49, January 2025. ISSN 3005-0677. doi: 10.1038/s44358-024-00005-w.
- [4] Maria Dornelas, Nicholas J. Gotelli, Brian McGill, Hideyasu Shimadzu, Faye Moyes, Caya Sievers, and Anne E. Magurran. Assemblage Time Series Reveal Biodiversity Change but Not Systematic Loss. *Science*, 344(6181):296–299, April 2014. ISSN 1095-9203. doi: 10.1126/science.1248484.
- [5] Barnabas H. Daru, T. Jonathan Davies, Charles G. Willis, Emily K. Meineke, Argo Ronk, Martin Zobel, Meelis Pärtel, Alexandre Antonelli, and Charles C. Davis. Widespread homogenization of plant communities in the Anthropocene. *Nature Communications*, 12(1), December 2021. ISSN 2041-1723. doi: 10.1038/s41467-021-27186-8.
- [6] Nate G. McDowell, Craig D. Allen, Kristina Anderson-Teixeira, Brian H. Aukema, Ben Bond-Lamberty, Louise Chini, James S. Clark, Michael Dietze, Charlotte Grossiord, Adam Hanbury-Brown, George C. Hurtt, Robert B. Jackson, Daniel J. Johnson, Lara Kueppers, Jeremy W. Lichstein, Kiona Ogle, Benjamin Poulter, Thomas A. M. Pugh, Rupert Seidl, Monica G. Turner, Maria Uriarte, Anthony P. Walker, and Chonggang Xu. Pervasive shifts in forest dynamics in a changing world. *Science*, 368(6494), May 2020. ISSN 1095-9203. doi: 10.1126/science.aaz9463.

- [7] Katharina Albrich, Werner Rammer, and Rupert Seidl. Climate change causes critical transitions and irreversible alterations of mountain forests. *Global Change Biology*, 26(7):4013–4027, May 2020. ISSN 1365-2486. doi: 10.1111/gcb.15118.
- [8] Giovanni Forzieri, Vasilis Dakos, Nate G. McDowell, Alkama Ramdane, and Alessandro Cescatti. Emerging signals of declining forest resilience under climate change. *Nature*, 608(7923):534–539, July 2022. ISSN 1476-4687. doi: 10.1038/s41586-022-04959-9.
- [9] Jessie J. Foest, Michał Bogdziewicz, Mario B. Pesendorfer, Davide Ascoli, Andrea Cutini, Anita Nussbaumer, Arne Verstraeten, Burkhard Beudert, Francesco Chianucci, Francesco Mezzavilla, Georg Gratzer, Georges Kunstler, Henning Meesenburg, Markus Wagner, Martina Mund, Nathalie Cools, Stanislav Vacek, Wolfgang Schmidt, Zdeněk Vacek, and Andrew Hacket-Pain. Widespread breakdown in masting in European beech due to rising summer temperatures. *Global Change Biology*, 30(5), May 2024. ISSN 1365-2486. doi: 10.1111/gcb.17307.
- [10] Andrew Hacket-Pain, Jakub Szymkowiak, Valentin Journé, Maciej K. Barczyk, Peter A. Thomas, Jonathan G. A. Lageard, Dave Kelly, and Michał Bogdziewicz. Growth decline in European beech associated with temperature-driven increase in reproductive allocation. *Proceedings of the National Academy of Sciences*, 122(5), January 2025. ISSN 1091-6490. doi: 10.1073/pnas.2423181122.
- [11] Debojyoti Chakraborty, Albert Ciceu, Dalibor Ballian, Marta Benito Garzón, Andreas Bolte, Gregor Bozic, Rafael Buchacher, Jaroslav Čepl, Eva Cremer, Alexis Ducousso, Julian Gaviria, Jan Peter George, André Hardtke, Mladen Ivankovic, Marcin Klisz, Jan Kowalczyk, Antoine Kremer, Milan Lstibůrek, Roman Longauer, Georgeta Mihai, László Nagy, Krasimira Petkova, Emil Popov, Randolph Schirmer, Tore Skrøppa, Thomas Mørtvedt Solvin, Arne Steffenrem, Jan Stejskal, Srdjan Stojnic, Katharina Volmer, and Silvio Schueler. Assisted tree migration can preserve the European forest carbon sink under climate change. *Nature Climate Change*, 14(8):845–852, 2024. ISSN 1758-6798. doi: 10.1038/s41558-024-02080-5.
- [12] Jessie J. Foest, Jakub Szymkowiak, Marcin K. Dyderski, Dave Kelly, Georges Kunstler, Szymon Jastrzębowski, and Michał Bogdziewicz. Forest tree fecundity declines as climate shifts. *Nature Climate Change*, 16(6):719–727, May 2026. ISSN 1758-6798. doi: 10.1038/s41558-026-02638-5.
- [13] Daniel H Janzen. Seeding patterns of tropical trees. In *Tropical trees as living systems*, pages 83–128. Cambridge University Press Cambridge, 1978.
- [14] Dave Kelly. The evolutionary ecology of mast seeding. *Trends in Ecology & Evolution*, 9(12): 465–470, December 1994. ISSN 0169-5347. doi: 10.1016/0169-5347(94)90310-7.
- [15] Michał Bogdziewicz, Dave Kelly, Davide Ascoli, Thomas Caignard, Francesco Chianucci, Elizabeth E. Crone, Emilie Fleurot, Jessie J. Foest, Georg Gratzer, Tomika Hagiwara, Qingmin

- Han, Valentin Journé, Léa Keurinck, Katarzyna Kondrat, Ryan McClory, Jalene M. LaMontagne, Ignacio A. Mundo, Anita Nussbaumer, Iris Oberklammer, Misuzu Ohno, Ian S. Pearse, Mario B. Pesendorfer, Giulia Resente, Akiko Satake, Mitsue Shibata, Rebecca S. Snell, Jakub Szymkowiak, Laura Touzot, Rafal Zwolak, Magdalena Zywiec, and Andrew J. Hacket-Pain. Evolutionary ecology of masting: mechanisms, models, and climate change. *Trends in Ecology & Evolution*, 39(9):851–862, September 2024. ISSN 0169-5347. doi: 10.1016/j.tree.2024.05.006.
- [16] Daniel H. Janzen. Seed Predation by Animals. *Annual Review of Ecology and Systematics*, 2(1):465–492, November 1971. ISSN 0066-4162. doi: 10.1146/annurev.es.02.110171.002341.
- [17] Stephanie M. Carlson, Curry J. Cunningham, and Peter A.H. Westley. Evolutionary rescue in a changing world. *Trends in Ecology & Evolution*, 29(9):521–530, September 2014. ISSN 0169-5347. doi: 10.1016/j.tree.2014.06.005.
- [18] Megan Bontrager and Amy L. Angert. Gene flow improves fitness at a range edge under climate change. *Evolution Letters*, 3(1):55–68, February 2019. ISSN 2056-3744. doi: 10.1002/evl3.91.
- [19] Dave Kelly and Victoria L. Sork. Mast Seeding in Perennial Plants: Why, How, Where? *Annual Review of Ecology and Systematics*, 33(1):427–447, November 2002. ISSN 0066-4162. doi: 10.1146/annurev.ecolsys.33.020602.095433.
- [20] Anita Nussbaumer, Peter Waldner, Vladislav Apuhtin, Fatih Aytar, Sue Benham, Filippo Bussotti, Johannes Eichhorn, Nadine Eickenscheidt, Petr Fabianek, Lutz Falkenried, Stefan Leca, Martti Lindgren, María José Manzano Serrano, Stefan Neagu, Seppo Nevalainen, Jozef Pajtik, Nenad Potočić, Pasi Rautio, Geert Sioen, Vidas Stakėnas, Celal Tasdemir, Iben Margrete Thomsen, Volkmar Timmermann, Liisa Ukonmaanaho, Arne Verstraeten, Sören Wulff, and Arthur Gessler. Impact of weather cues and resource dynamics on mast occurrence in the main forest tree species in Europe. *Forest Ecology and Management*, 429:336–350, December 2018. ISSN 0378-1127. doi: 10.1016/j.foreco.2018.07.011.
- [21] Valentin Journé, Dave Kelly, Andrew Hacket-Pain, Ian S. Pearse, Jakub Szymkowiak, Jessie J. Foest, Katarzyna Kondrat, Iris Oberklammer, Mario B. Pesendorfer, Akiko Satake, and Michał Bogdziewicz. Weather drivers of reproductive variability in perennial plants and their implications for climate change risks. *Nature Communications*, 16(1), October 2025. ISSN 2041-1723. doi: 10.1038/s41467-025-64300-6.
- [22] Giorgio Vacchiano, Andrew Hacket-Pain, Marco Turco, Renzo Motta, Janet Maringer, Marco Conedera, Igor Drobyshev, and Davide Ascoli. Spatial patterns and broad-scale weather cues of beech mast seeding in Europe. *New Phytologist*, 215(2):595–608, May 2017. ISSN 1469-8137. doi: 10.1111/nph.14600.
- [23] Juan Gallego Zamorano, Tatu Hokkanen, and Aleksi Lehikoinen. Climate-driven synchrony in seed production of masting deciduous and conifer tree species. *Journal of Plant Ecology*, page rtw117, October 2016. ISSN 1752-993X. doi: 10.1093/jpe/rtw117.

- [24] C. M. Zohner, L. D. Mo, S. S. Renner, J. C. Svenning, Y. Vitasse, B. M. Benito, A. Ordonez, F. Baumgarten, J. F. Bastin, V. Sebold, P. B. Reich, J. J. Liang, G. J. Nabuurs, S. de Miguel, G. Alberti, C. Anton-Fernandez, R. Balazy, U. B. Brandli, H. Y. H. Chen, C. Chisholm, E. Cienciala, S. Dayanandan, T. M. Fayle, L. Frizzera, D. Gianelle, A. M. Jagodzinski, B. Jaroszewicz, T. Jucker, S. Kepfer-Rojas, M. L. Khan, H. S. Kim, H. Korjus, V. K. Johannsen, D. Laarmann, M. Lang, T. Zawila-Niedzwiecki, P. A. Niklaus, A. Paquette, H. Pretzsch, P. Saikia, P. Schall, V. Seben, M. Svoboda, E. Tikhonova, H. Viana, C. Y. Zhang, X. H. Zhao, and T. W. Crowther. Late-spring frost risk between 1959 and 2017 decreased in North America but increased in Europe and Asia. *Proceedings of the National Academy of Sciences of the United States of America*, 117(22):12192–12200, 2020.
- [25] C. J. Chamberlain, B. I. Cook, I. Morales-Castilla, and E. M. Wolkovich. Climate change reshapes the drivers of false spring risk across European trees. *New Phytologist*, 229(1):323–334, 2021.
- [26] Dominik L. Schumacher, Jitendra Singh, Mathias Hauser, Erich M. Fischer, Martin Wild, and Sonia I. Seneviratne. Exacerbated summer European warming not captured by climate models neglecting long-term aerosol changes. *Communications Earth & Environment*, 5(1), April 2024. ISSN 2662-4435. doi: 10.1038/s43247-024-01332-8.
- [27] Michał Bogdziewicz, Dave Kelly, Andrew J. Tanentzap, Peter Thomas, Jessie Foest, Jonathan Lageard, and Andrew Hacket-Pain. Reproductive collapse in European beech results from declining pollination efficiency in large trees. *Global Change Biology*, 29(16):4595–4604, May 2023. ISSN 1365-2486. doi: 10.1111/gcb.16730.
- [28] Monica A. Geber, Maxine A. Watson, and Hans de Kroon. *Organ Preformation, Development, and Resource Allocation in Perennials*, pages 113–141. Elsevier, 1997. ISBN 9780120834907. doi: 10.1016/b978-012083490-7/50006-2.
- [29] J. D. Wilkie, M. Sedgley, and T. Olesen. Regulation of floral initiation in horticultural trees. *Journal of Experimental Botany*, 59(12):3215–3228, 2008. ISSN 1460-2431. doi: 10.1093/jxb/ern188.
- [30] S.P. Monselise and E.E. Goldschmidt. Alternate Bearing in Fruit Trees. *Horticultural Reviews*, pages 128–173, January 1982. doi: 10.1002/9781118060773.ch5.
- [31] K.F. Bangerth. Floral induction in mature, perennial angiosperm fruit trees: Similarities and discrepancies with annual/biennial plants and the involvement of plant hormones. *Scientia Horticulturae*, 122(2):153–163, 2009. ISSN 0304-4238. doi: 10.1016/j.scienta.2009.06.014.
- [32] Anton Milyaev, Julian Kofler, Yudelsy Antonia Tandron Moya, Janne Lempe, Dario Stefanelli, Magda-Viola Hanke, Henryk Flachowsky, Nicolaus von Wirén, and Jens-Norbert Wünsche. Profiling of phytohormones in apple fruit and buds regarding their role as potential regulators

of flower bud formation. *Tree Physiology*, 42(11):2319–2335, August 2022. ISSN 1758-4469. doi: 10.1093/treephys/tpac083.

[33] Valentin Journé, Jakub Szymkowiak, Jessie Foest, Andrew Hacket-Pain, Dave Kelly, and Michał Bogdziewicz. Summer solstice orchestrates the subcontinental-scale synchrony of mast seeding. *Nature Plants*, 10(3):367–373, March 2024. ISSN 2055-0278. doi: 10.1038/s41477-024-01651-w.

[34] Carol K. Augspurger. Spring 2007 warmth and frost: phenology, damage and refoliation in a temperate deciduous forest. *Functional Ecology*, 23(6):1031–1039, November 2009. ISSN 1365-2435. doi: 10.1111/j.1365-2435.2009.01587.x.

[35] John R. Packham, Peter A. Thomas, Mark D. Atkinson, and Thomas Degen. Biological Flora of the British Isles: *Fagus sylvatica*. *Journal of Ecology*, 100(6):1557–1608, October 2012. ISSN 1365-2745. doi: 10.1111/j.1365-2745.2012.02017.x.

[36] Yann Vitasse and Martine Rebetez. Unprecedented risk of spring frost damage in Switzerland and Germany in 2017. *Climatic Change*, 149(2):233–246, July 2018. ISSN 1573-1480. doi: 10.1007/s10584-018-2234-y.

[37] Michał Bogdziewicz, Andrew Hacket-Pain, Davide Ascoli, and Jakub Szymkowiak. Environmental variation drives continental-scale synchrony of European beech reproduction. *Ecology*, 102(7), 2021. ISSN 1939-9170. doi: 10.1002/ecy.3384.

[38] Walter D. Koenig and Johannes M. H. Knops. Scale of mast-seeding and tree-ring growth. *Nature*, 396(6708):225–226, November 1998. ISSN 1476-4687. doi: 10.1038/24293.

[39] Jalene M. LaMontagne, Ian S. Pearse, David F. Greene, and Walter D. Koenig. Mast seeding patterns are asynchronous at a continental scale. *Nature Plants*, 6(5):460–465, April 2020. ISSN 2055-0278. doi: 10.1038/s41477-020-0647-x.

[40] T. Jonathan Davies and Ailene MacPherson. Seed masting as a mechanism for escape from pathogens. *Current Biology*, 34(4):R120–R125, February 2024. ISSN 0960-9822. doi: 10.1016/j.cub.2023.12.027.

[41] Frederik Sachser, Georg Gratzner, Mario B. Pesendorfer, Heino Konrad, Iris Kempter, and Ursula Nopp-Mayr. Tracking individual seed fate confirms mainly antagonistic interactions between rodents and European beech. *Biology Letters*, 21(1), January 2025. ISSN 1744-957X. doi: 10.1098/rsbl.2024.0586.

[42] Dave Kelly, Andre Geldenhuis, Alex James, E. Penelope Holland, Michael J. Plank, Robert E. Brockie, Philip E. Cowan, Grant A. Harper, William G. Lee, Matt J. Maitland, Alan F. Mark, James A. Mills, Peter R. Wilson, and Andrea E. Byrom. Of mast and mean: differential-temperature cue makes mast seeding insensitive to climate change. *Ecology Letters*, 16(1): 90–98, November 2012. ISSN 1461-0248. doi: 10.1111/ele.12020.

- [43] Michał Bogdziewicz, Andrew Hacket-Pain, Dave Kelly, Peter A. Thomas, Jonathan Lageard, and Andrew J. Tanentzap. Climate warming causes mast seeding to break down by reducing sensitivity to weather cues. *Global Change Biology*, 27(9):1952–1961, March 2021. ISSN 1365-2486. doi: 10.1111/gcb.15560.
- [44] Stephen C. Stearns. *The Evolution of Life Histories*. Oxford University Press, Oxford, UK, 1992. ISBN 9780198577416.
- [45] Michał Bogdziewicz, Dave Kelly, Peter A. Thomas, Jonathan G. A. Lageard, and Andrew Hacket-Pain. Climate warming disrupts mast seeding and its fitness benefits in European beech. *Nature Plants*, 6(2):88–94, February 2020. ISSN 2055-0278. doi: 10.1038/s41477-020-0592-8.
- [46] Anita Nussbaumer, Katrin Meusburger, Maria Schmitt, Peter Waldner, Regula Gehrig, Matthias Haeni, Andreas Rigling, Ivano Brunner, and Anne Thimonier. Extreme summer heat and drought lead to early fruit abortion in European beech. *Scientific Reports*, 10(1), March 2020. ISSN 2045-2322. doi: 10.1038/s41598-020-62073-0.
- [47] M.D. Petrie, R.M. Hubbard, J.B. Bradford, T.E. Kolb, A. Noel, D.R. Schlaepfer, M.A. Bowen, L.R. Fuller, and W.K. Moser. Widespread regeneration failure in ponderosa pine forests of the southwestern United States. *Forest Ecology and Management*, 545:121208, October 2023. ISSN 0378-1127. doi: 10.1016/j.foreco.2023.121208.
- [48] Geoffrey M. Hilton and John R. Packham. Annual and regional variation in English beech mast (*Fagus Sylvatica* L.). *Arboricultural Journal*, 10(1):3–14, February 1986. ISSN 2168-1074. doi: 10.1080/03071375.1986.9756322.
- [49] J. R. Packham, P. A. Thomas, J. G.A. Lageard, and G. M. Hilton. The English beech masting survey 1980–2007: variation in the fruiting of the common beech (*Fagus Sylvatica* L.) and its effects on woodland ecosystems. *Arboricultural Journal*, 31(3):189–214, October 2008. ISSN 2168-1074. doi: 10.1080/03071375.2008.9747535.
- [50] Joaquín Muñoz-Sabater, Emanuel Dutra, Anna Agustí-Panareda, Clément Albergel, Gabriele Arduini, Gianpaolo Balsamo, Souhail Boussetta, Margarita Choulga, Shaun Harrigan, Hans Hersbach, Brecht Martens, Diego G. Miralles, María Piles, Nemesio J. Rodríguez-Fernández, Ervin Zsoter, Carlo Buontempo, and Jean-Noël Thépaut. ERA5-Land: a state-of-the-art global reanalysis dataset for land applications. *Earth System Science Data*, 13(9):4349–4383, 2021. ISSN 1866-3516. doi: 10.5194/essd-13-4349-2021.
- [51] Bob Carpenter, Andrew Gelman, Matthew D. Hoffman, Daniel Lee, Ben Goodrich, Michael Betancourt, Marcus Brubaker, Jiqiang Guo, Peter Li, and Allen Riddell. Stan: A Probabilistic Programming Language. *Journal of Statistical Software*, 76(1), 2017. ISSN 1548-7660. doi: 10.18637/jss.v076.i01.